End-of-year and oral workshops

O1 · Turning a theme into a question

S82 · 2 h. Choose a situation that allows reasoning from the syllabus, then write a precise question. Examples: “How do we choose a number of trials to make a failure rare?”; “How can a cooling model be used to predict a waiting time?”; “Why does a mean become more stable as the sample grows?” Produce a page containing the question, the hypotheses, a numerical example, an accessible proof and a limitation of the model. Allow 20 min of individual research, 30 min of mathematical work, 30 min of group discussion, 25 min of writing and 15 min of feedback.

O2 · Presenting and answering

S84 · 2 h. Prepare a 10 min presentation and answers to the following questions: why this model? which hypothesis is decisive? what does each variable mean? which result is proved? what changes with another value of the parameter? Practise in groups of three: a speaker, an interviewer and an observer. The observer notes one clear step, one step to justify and one wording to improve. Swap roles. A video is useful only if it is permitted and actually watched back.

O3 · Oral simulation

S87 and S93. Prepare two questions and give a presentation followed by a discussion. Use a simple grid: mathematical accuracy, structure, explanation of notation, response to objections, quality of the exchange. The final arrangements are those of the instructions and summons for the session.